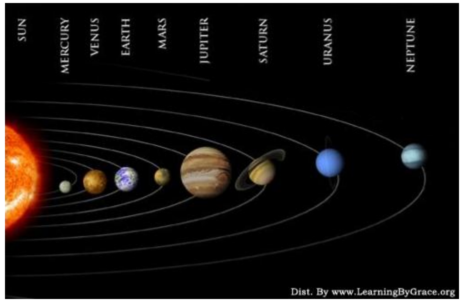
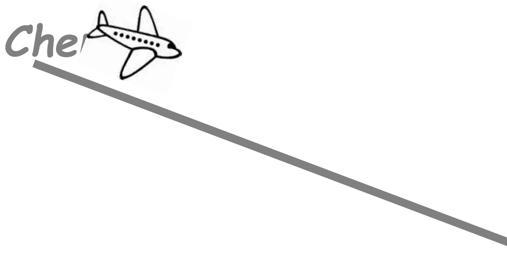
Chapter 2 &3
Motion in One, Two and Three Dimensions

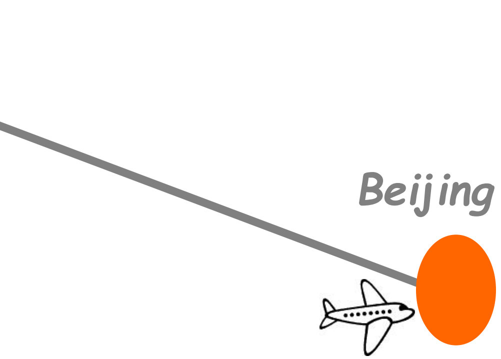
- Position and Displacement
- Velocity
- Acceleration
- 2D and 3D Motion

Motion

- Everything moves! Motion is one of the main topics in Mechanics.
- In the spirit of taking things apart for study, then putting them back together, we will first consider only motion along a straight line.
- Simplification: Consider a moving object as a particle, i.e. it moves like a particle—a "point object"







Prepare Knowledge

Particle Model / Point Object

Condition:

- If we study the motion of an object, we can ignore the effect of its size and shape on the motion of the object.
- If we do not involve the rotation and deformation of the object, we can treat the object as a point with mass (ie, mass point). .

Definition:

- we can treat the object as a point with mass (ie, mass point)
- Normally we say particle

Reference Frames and Displacement

The world, and everything in it, moves. Even seemingly stationary things, such as a house moves with the Earth, the Earth's orbit around the Sun, the Sun's orbit around the center of the Milky Way galaxy, and that galaxy's migration relative to other galaxies.

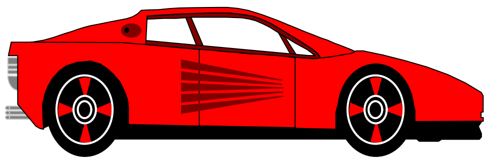
- **Absolutely stationary objects are not available.**
- **Motion is absolute.**

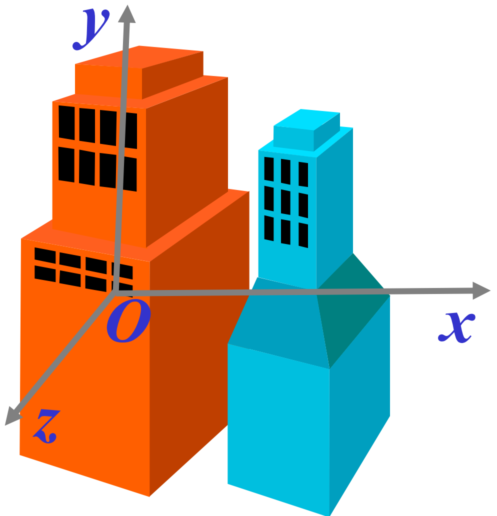
- **The description of the movement is relative.**

To describe the position of an object, the other object which is referred to (Reference Frame) should be chosen. It is arbitrary.

To determine the location of a body at the reference object quantitatively, a coordinate system is built on it.

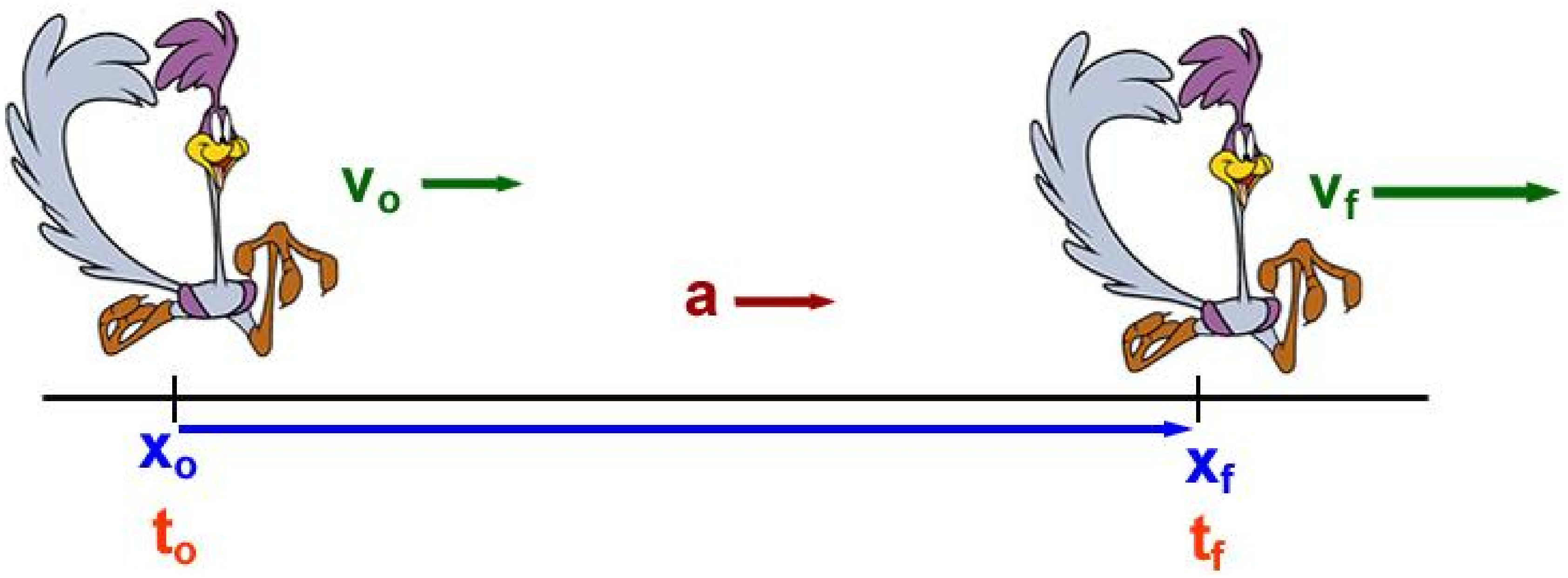
An observational reference frame , often referred to as a physical frame of reference, a frame of reference, or simply a frame, is a *physical concept related to an observer and the observer's state of motion.*





Basic Quantities in Kinematics

Displacement*, *Velocity*, *Time* and *Acceleration

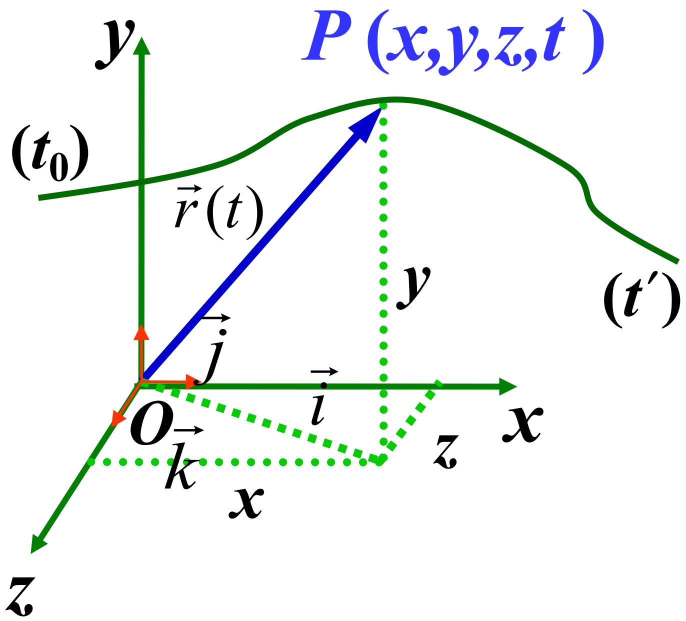


Quantities in Motion

- Any motion involves three concepts
  - **Displacement**
  - **Velocity**
  - **Acceleration**
- These concepts can be used to study objects in motion.

Vector Kinematics

1. Position vector––The location of a particle relative to the origin of a coordinate system.



For a Cartesian system:

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

Motion function :

$$\vec{r} = \vec{r}(t)$$

$$= x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$$

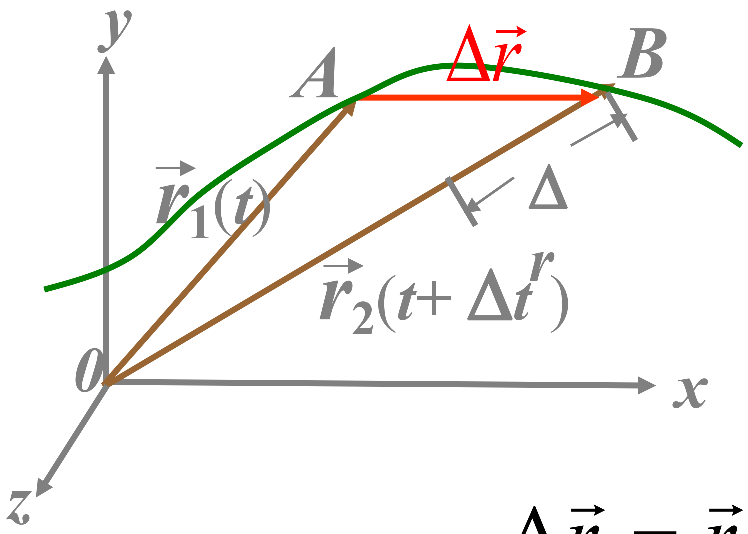
Trajectory equation :

$$f(x, y, z) = 0$$

How to transform the motion equation into the trajectory equation?

Motion function $\xrightarrow{\textbf{eliminate } t}$ **Trajectory equation**

2. Displacement-- $\Delta\vec{r}$ (A particle is changing in its position)



The displacement vector extends from the head of the initial position vector to the head of the later position vector.

$$\Delta\vec{r} = \vec{r}_2(t+\Delta t) - \vec{r}_1(t) = \vec{r}_2 - \vec{r}_1$$

$$= (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} + (z_2 - z_1)\vec{k}$$

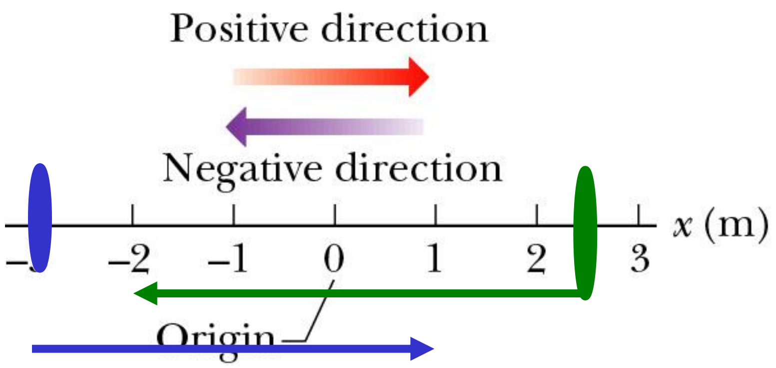
$$= \Delta x\vec{i} + \Delta y\vec{j} + \Delta z\vec{k}$$

Notes about displacement:

(i) It is a vector quantity —— The magnitude of vector should be the length of this vector, i.e.

$$|\Delta\vec{r}|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

- It has both magnitude and direction: + or - sign
- It has units of [length]: meters.



$x_1(t_1) = +2.5$ m
$x_2(t_2) = -2.0$ m
$\Delta x = -2.0$ m $- 2.5$ m $= -4.5$ m

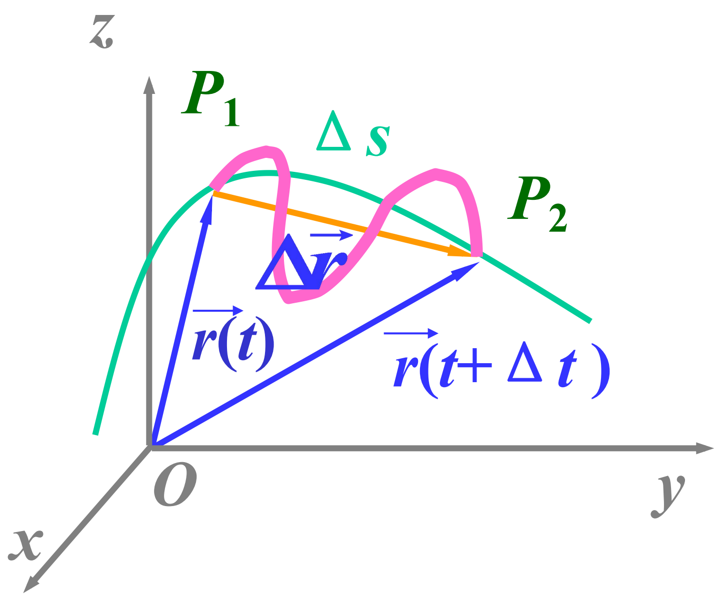
$x_1(t_1) = -3.0$ m
$x_2(t_2) = +1.0$ m
$\Delta x = +1.0$ m $+ 3.0$ m $= +4.0$ m

Notes about displacement:

(ii) Different from the path ΔS

Path is the total lengths of the path curve， scalar.

Displacement depends only on the position of head to tail, independent with the real path .



Hey! $\left|\Delta \vec{r}\right| \neq \Delta s$

In the limit as $\Delta t \to 0$, $\boxed{\left|\mathrm{d}\,\vec{r}\right| = \left|\mathrm{d}\,s\right|}$

(iii) The displacement is independent on the choice of origin.

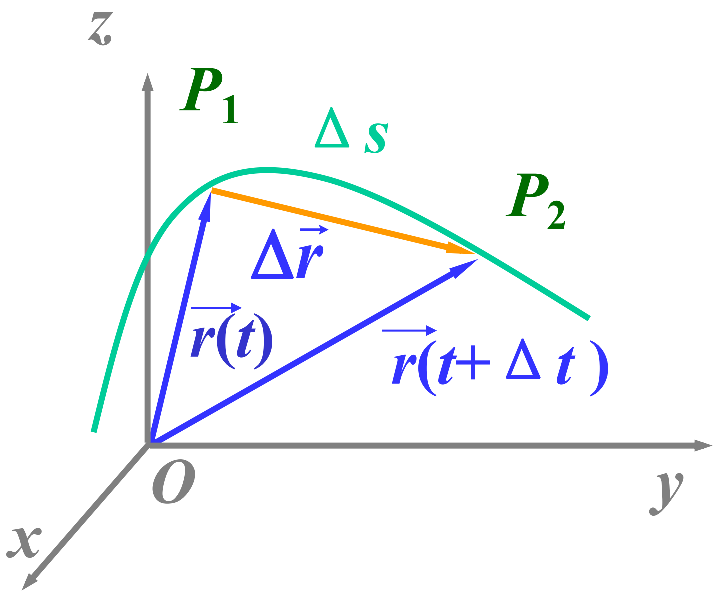
3. Average Velocity and Instantaneous Velocity

3.1. Average velocity

$$\text{average velocity} = \frac{\text{displacement}}{\text{time elapsed}}$$

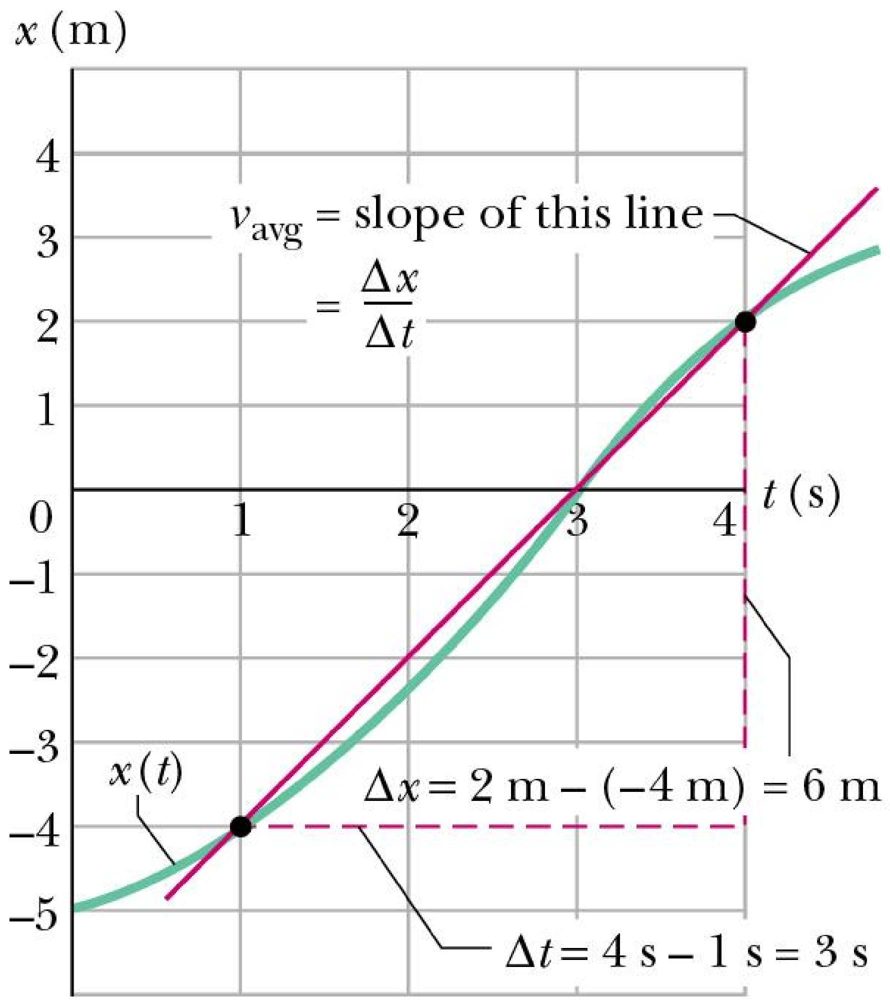
(It is defined as the total displacement traveled divided by the time it takes to travel this displacement)

$$\bar{\vec{v}} = \frac{\Delta \vec{r}}{\Delta t}$$



$$\begin{cases} \text{Direction} \quad \text{is along} \quad \Delta\vec{r} \\ \text{Magnitude} \quad \text{is } |\bar{\vec{v}}| \end{cases}$$

3.1. Average velocity

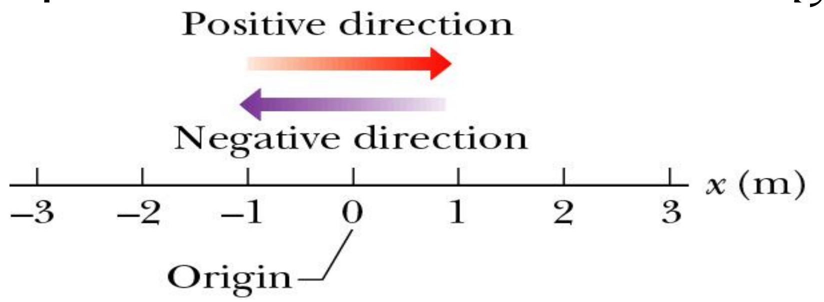


- Average velocity

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$

is the slope of the line segment between end points on a graph.

- Dimensions: length/time (L/T) [m/s].
- SI unit: m/s.
- It is a vector (i.e. is signed), and displacement direction sets its sign.



Example 1（P29）

Table 2.1 Position of the Car at Various Times

Position	t (s)	x (m)
Ⓐ	0	30
Ⓑ	10	52
Ⓒ	20	38
Ⓓ	30	0
Ⓔ	40	−37
Ⓕ	50	−53

As an example, we can use the data in Table 2.1 to find the average velocity in the time interval from point Ⓐ to point Ⓑ (assume two digits are significant):

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{52\text{ m} - 30\text{ m}}{10\text{ s} - 0\text{ s}} = 2.2\text{ m/s}$$

And how about from point Ⓒ to point Ⓓ

The average velocity of an object in one dimension can be either positive or negative, depending on the sign of the displacement. (The time interval Δt is always positive.)

3.2. Instantaneous velocity

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t}$$

Δt gets extremely close to zero, Called the derivative of $\vec{r}$ with respect to t

$$= \frac{\mathrm{d}}{\mathrm{d}t}\left[x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}\right]$$

$$\boxed{= \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k}}$$

$$\begin{cases} \text{Direction is along tangent line} \\ \text{Magnitude is } v = |\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2} \end{cases}$$

Notes about Instantaneous Velocity:

- Instantaneous means "at some given instant". The instantaneous velocity indicates what is happening at every point of time.
- Limiting process:
  - Chords approach the tangent as $\Delta t \Rightarrow 0$
  - Slope measure rate of change of position
- It is a vector quantity.
- Dimension: length/time (L/T), [m/s].
- It is the slope of the tangent line to x(t).
- Instantaneous velocity v(t) is a function of time.

Example

A particle with a motion function $\vec{r} = 3t^2\vec{i} + 2t\vec{j} + 4\vec{k}$.

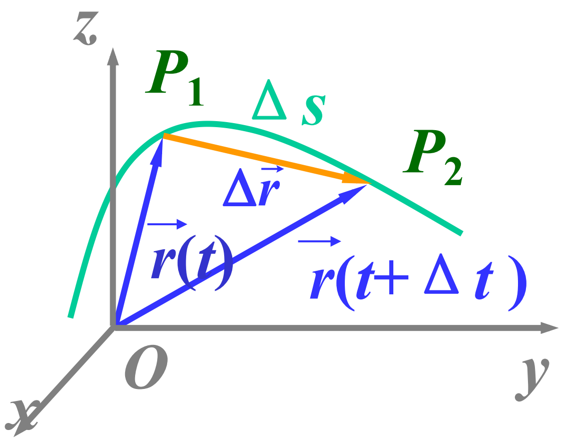
What is the velocity at t=2s?

3.3. Average and Instantaneous speed

$$\text{average speed} = \frac{\text{distance travel}}{\text{time elapsed}}$$

(It is defined as the total distance traveled along its path divided by the time it takes to travel this distance)

Average speed: $\bar{v} = \dfrac{\Delta s}{\Delta t}$



Instantaneous speed:

$$v = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

Called the derivative of s with respect to t

Notes about Average and Instantaneous speed

- Dimension: length/time, [m/s]
- Scalar: No direction involved.

3.4. The relation between the velocity and speed

$$\because \Delta t \to 0 \qquad |\mathrm{d}\vec{r}| = \mathrm{d}s \qquad \therefore v = \frac{ds}{dt} = \left|\frac{d\vec{r}}{dt}\right| = |\vec{v}|$$

The magnitude of velocity equals to speed .

Note:

Speed: $\boxed{v = \frac{\mathrm{d}s}{\mathrm{d}t}}$

Speed is a scalar.

Example

2.1 Figure 2.4 shows the unusual path of a confused football player. After receiving a kickoff at his own goal, he runs downfield to within inches of a touchdown, then reverses direction and races back until he's tackled at the exact location where he first caught the ball. During this run, which took 25 s, what is (a) the path length he travels, (b) his displacement, and (c) his average velocity in the x-direction? (d) What is his average speed?

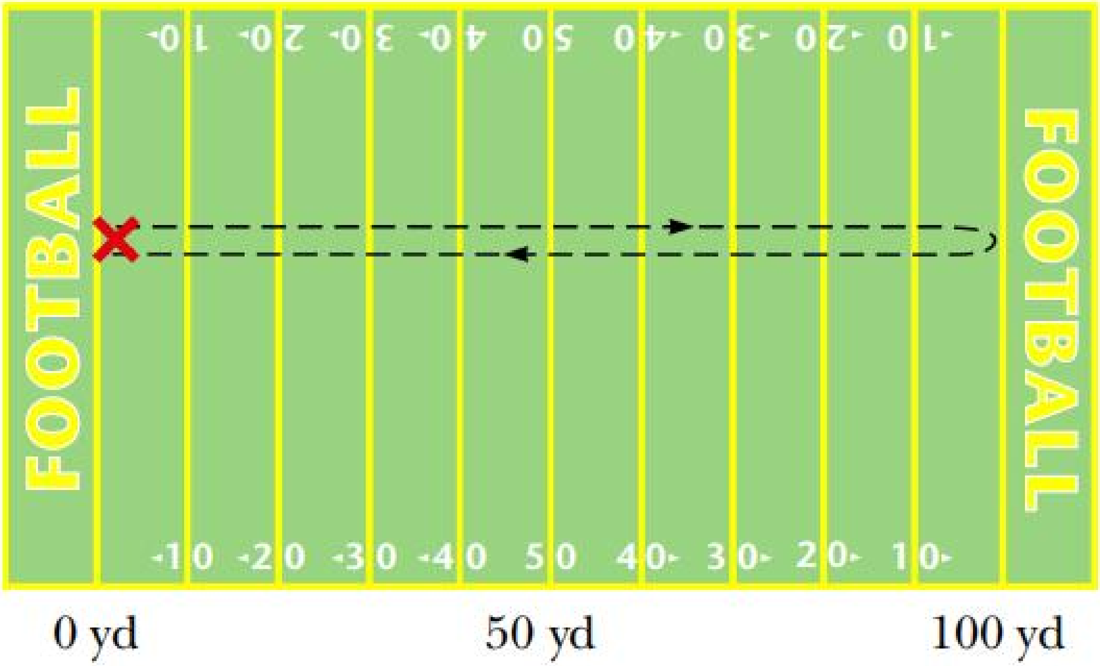


Figure 2.4 (Quick Quiz 2.1) The path followed by a confused football player.

4. Acceleration

Going from place to place in your car, you rarely travel long distances at constant velocity. The velocity of the car increases when you step harder on the gas pedal and decreases when you apply the brakes. The velocity also changes when you round a curve, altering your direction of motion. The changing of an object's velocity with time is called acceleration.

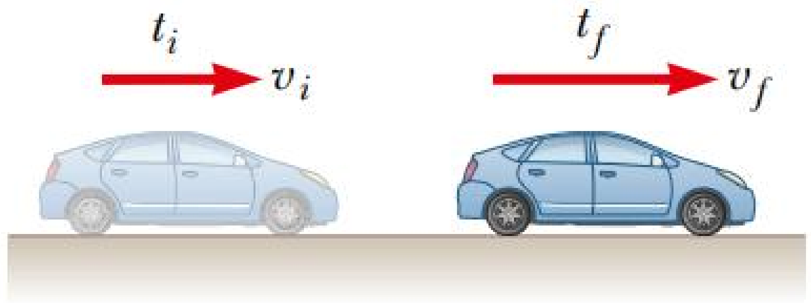


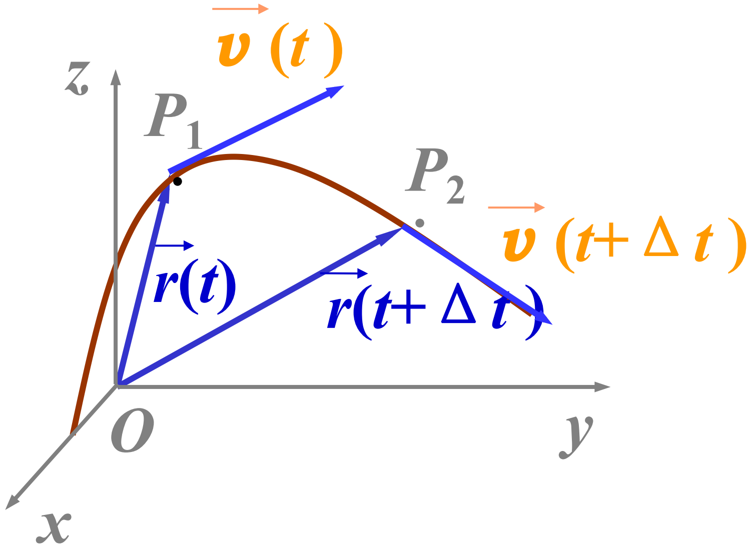
Figure 2.8 A car moving to the right accelerates from a velocity of v_i to a velocity of v_f in the time interval $\Delta t = t_f - t_i$.

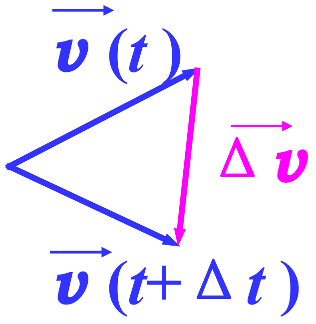
4. Acceleration

4.1 Average acceleration:

It is defined as the change of the velocity divided by the time taken to make this change.

$$\overline{\vec{a}} = \frac{\Delta \vec{v}}{\Delta t}$$





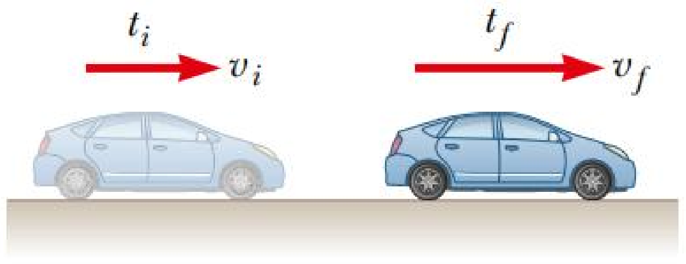


Figure 2.8 A car moving to the right accelerates from a velocity of v_i to a velocity of v_f in the time interval $\Delta t = t_f - t_i$.

For example, suppose the car shown in Figure 2.8 accelerates from an initial velocity of v_i =10 m/s to a final velocity of v_f =20 m/s in a time interval of 2 s. (Both velocities are toward the right, selected as the positive direction.)We can find the average acceleration:

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{20\ \mathrm{m/s} - 10\ \mathrm{m/s}}{2\ \mathrm{s}} = +5\ \mathrm{m/s^2}$$

Suppose the velocity of a car changes from -10 m/s to -20 m/s in a time interval of 2 s.

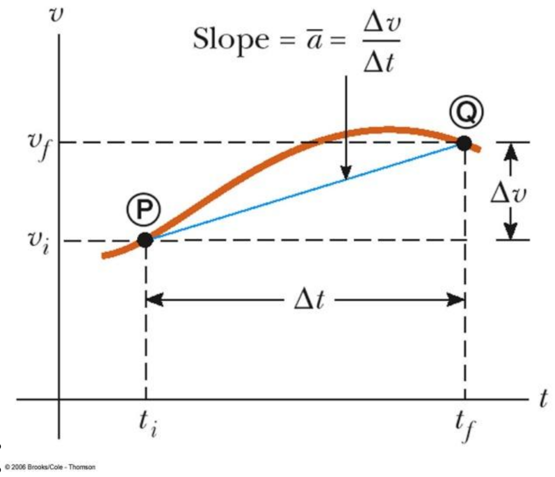
How about its average acceleration?

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{-20\ \mathrm{m/s} - (-10\ \mathrm{m/s})}{2\ \mathrm{s}} = -5\ \mathrm{m/s^2}$$

The minus sign indicates that the acceleration vector is also in the negative x-direction. Because the velocity and acceleration vectors are in the same direction, the speed of the car must increase as the car moves to the left. Positive and negative accelerations specify directions relative to chosen axes, not "speeding up" or "slowing down." The terms speeding up or slowing down refer to an increase and a decrease in speed, respectively.

Notes about Average Acceleration:

- Changing velocity (non-uniform) means an acceleration is present.
- Acceleration is the rate of change of velocity.
- Acceleration is a vector quantity.
- Acceleration has both magnitude and direction.
- Acceleration has a dimensions of length/time^2: [m/s^2].
- It is tempting to call a negative acceleration a "deceleration," but note:
  - When the sign of the velocity and the acceleration are the same (either positive or negative), then the speed is increasing
  - When the sign of the velocity and the acceleration are in the opposite directions, the speed is decreasing
- Average acceleration is the slope of the line connecting the initial and final velocities on a velocity-time graph.



4.2 Instantaneous acceleration:

It is defined as the limiting value of the average acceleration as we let Δt approach zero.

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

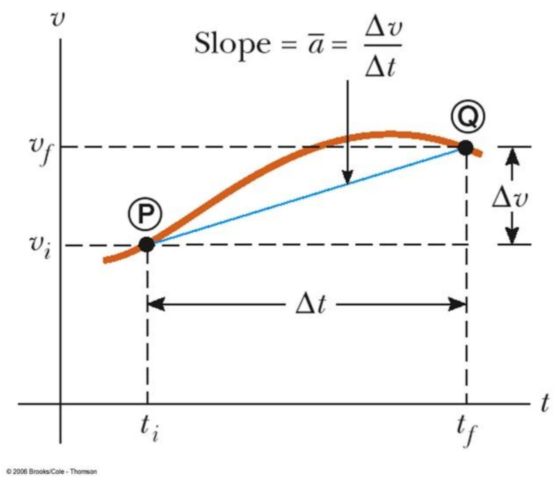
Is the derivative of $\vec{v}$ with respect to t

$$= \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d^2\vec{r}}{dt^2}$$

Is the second derivative of $\vec{r}$ with respect to t

Notes about Instantaneous acceleration:

- When the instantaneous accelerations are always the same, the acceleration will be uniform. The instantaneous acceleration will be equal to the average acceleration
- Instantaneous acceleration is the slope of the tangent to the curve of the velocity-time graph.



The component form in Cartesian coordinate is:

$$\vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

$$\begin{cases} a_x = \dfrac{\mathrm{d}v_x}{\mathrm{d}t} = \dfrac{\mathrm{d}^2 x}{\mathrm{d}t^2} \\ a_y = \dfrac{\mathrm{d}v_y}{\mathrm{d}t} = \dfrac{\mathrm{d}^2 y}{\mathrm{d}t^2} \\ a_z = \dfrac{\mathrm{d}v_z}{\mathrm{d}t} = \dfrac{\mathrm{d}^2 z}{\mathrm{d}t^2} \end{cases}$$

Is the derivative of v_x with respect to t

Direction of $\vec{a}$: The limit direction of $\mathrm{d}\vec{v}$

Magnitude of $\vec{a}$ $a = |\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$

Example

Acceleration given x(t). A particle is moving in a straight line so that its position is given by the relation $x = (2.10\,m/s^2)t^2 + (2.80m)$, as in Example 2-3. Calculate (a) its average acceleration during the time interval from $t_1 = 3.00s$ to $t_2 = 5.00s$, and (b) its instantaneous acceleration as a function of time.

Two kinds of questions belong to the Kinematics:

(i) From $\vec{r} \rightarrow \vec{v}$, or $\vec{a}$;

(ii) From $\vec{a}$ (or $\vec{v}$) $\rightarrow \vec{v}$, (or $\vec{r}$);

Example:

A cat runs across a parking lot. The coordinates of the cat's position as functions of time t are given by

$$x = -0.31\,t^2 + 7.2\,t + 28\,,\quad y = 0.22\,t^2 - 9.1\,t + 30\ \text{(SI)}$$

(a) At $t = 15$s, what is the cat's position vector r in unit-vector notation ?

(b) How about its velocity ; and (c) acceleration ?

Solution:

(a) $\vec{r} = (66.25\vec{i} - 57\vec{j})\text{m};\quad \boldsymbol{r = 87.40\ \text{m}}$

(b) $\vec{v} = (-2.1\vec{i} - 2.5\vec{j})$**m/s ; v =3.3 m/s**

(c) $\vec{a} = (-0.62\vec{i} + 0.44\vec{j})$**m/s^2 ; a =0.76 m/s^2**

Example:

A particle moves along the direction of x axis, $a=2t$. At $t=0$, $x0=0$， $v0=0$.

What are its velocity and position at $t=2$s?

Solution:

Acceleration, $a=2t$, is not constant.

$$\mathrm{d}v = 2t\mathrm{d}t$$

Simultaneous integration of both sides of equality

$$\int_0^v \mathrm{d}v = \int_0^t 2t\mathrm{d}t \quad ; \qquad v = \frac{\mathrm{d}x}{\mathrm{d}t} = t^2 \qquad (1)$$

$$\mathrm{d}x = t^2\mathrm{d}t \quad ; \quad \int_0^x \mathrm{d}x = \int_0^t t^2\mathrm{d}t \qquad ; \quad x = \frac{1}{3}t^3 \qquad (2)$$

So $\quad v = 4\ \mathrm{m/s}\ ; \qquad x = 8/3 = 2.67\ \mathrm{m}$

Example:

A particle moves along x direction, $a=2+6x^2$. At $t=0$, $x_0=0$, $v_0=0$. What is its $v(x)$?

Solution:

$$a=\frac{\mathrm{d}v}{\mathrm{d}t} \quad ;$$

$$\boldsymbol{a}=\frac{\mathbf{d}\boldsymbol{v}}{\mathbf{d}\boldsymbol{t}}=\frac{\mathbf{d}\boldsymbol{v}}{\mathbf{d}\boldsymbol{x}}\cdot\frac{\mathrm{d}x}{\mathbf{d}\boldsymbol{t}}=\boxed{\boldsymbol{v}\frac{\mathbf{d}\boldsymbol{v}}{\mathbf{d}\boldsymbol{x}}=(2+6x^2)}$$

Simultaneous integration of both sides of equality

$$\int_0^x(2+6x^2)\mathrm{d}x=\int_0^v v\mathrm{d}v$$

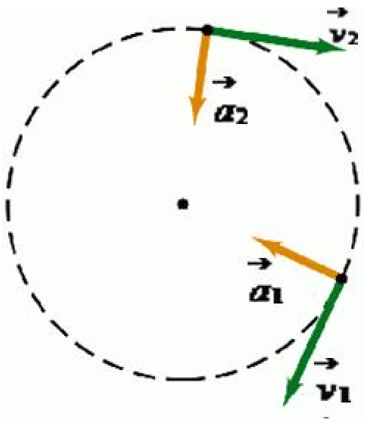
$$2(x+x^3)=\frac{1}{2}v^2$$

$$v=2\sqrt{x+x^3} \quad (\text{“}-\text{”} \quad \text{no physical meaning})$$

5.Circular Motion

5.1. Uniform circular motion

A particle travels around a circular path at a constant speed v, there is still an acceleration and it is always directed radially inward, centripetal



$$\vec{a} = \frac{v^2}{r}\hat{r}$$

$$\hat{r} = \frac{\vec{r}}{|\vec{r}\,|} = \frac{\vec{r}}{r}$$ (Radial Unit Vector)

The time for the particle to complete a circle is:

$$\boxed{T = \frac{2\pi r}{v}}$$ — period

Example

Acceleration of a revolving ball. A 150-g ball at the end of a string is revolving uniformly in a horizontal circle of radius 0.6 m. The ball makes 2.00 revolutions in a second. What is its centripetal acceleration?

Solution:

The centripetal acceleration is $a_R = v^2/r$ .First wo determine the speed of the ball v. The ball makes two complete revolutions per second. so its period is T = 0.5s In this time it travels one circumference of the circle, 2πr, so the ball has speed

$$v = \frac{2\pi r}{T} = \frac{2 \times 3.14 \times 0.6}{0.5} = 7.54\mathrm{m/s}$$

The centripetal acceleration is

$$a_R = \frac{v^2}{r} = \frac{7.54^2}{0.6} = 94.8\mathrm{m/s^2}$$

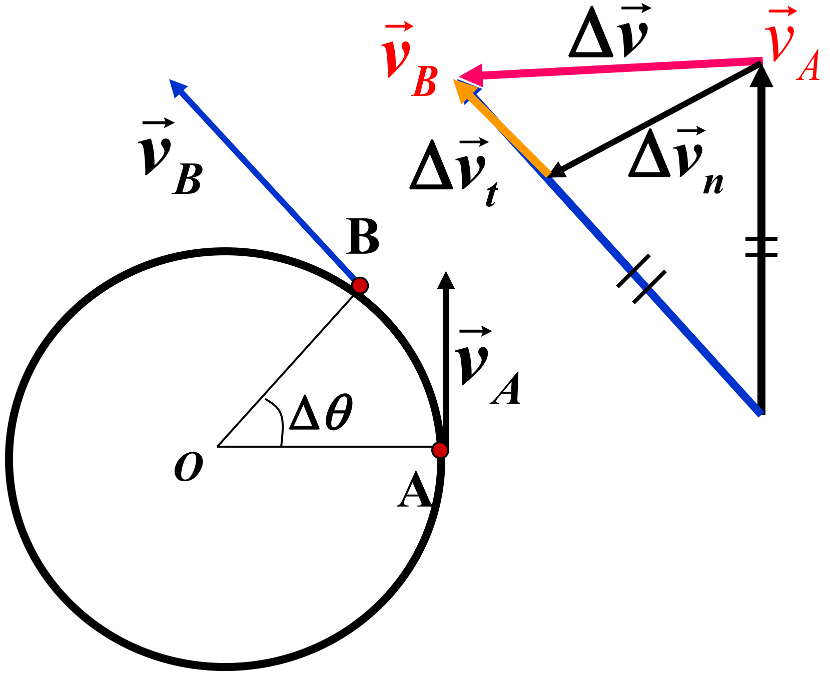
5.2 Nonuniform Circular Motion

A particle travels around a circular path, but both magnitude and direction of velocity change.

$$\Delta\vec{v} = \vec{v}_B - \vec{v}_A = \Delta\vec{v}_n + \Delta\vec{v}_t$$

$$\vec{a} = \lim_{\Delta t\to 0}\frac{\Delta\vec{v}}{\Delta t} = \lim_{\Delta t\to 0}\frac{\Delta\vec{v}_n}{\Delta t} + \lim_{\Delta t\to 0}\frac{\Delta\vec{v}_t}{\Delta t} = a_n\vec{n} + a_t\vec{\tau}$$

$$\boxed{\therefore\ \vec{a} = a_n\vec{n} + a_t\vec{\tau}}$$



$\Delta\vec{v}_t$: Changes in **magnitude**

$\Delta\vec{v}_n$: Changes in **direction**

(1)Tangential acceleration:

$$a_t = |\vec{a}_t| = \frac{\mathrm{d}v}{\mathrm{d}t}$$

It only indicates the change in magnitude of velocity。

The direction of $\vec{\boldsymbol{a}}_{\boldsymbol{t}}$: Tangential line.

(2)Normal acceleration：

$$|\vec{a}_n| = \frac{v^2}{r}$$

It only indicates the change in direction of velocity。

The direction of $\vec{a}_n$: radial direction.

a_n, normal acceleration is also called a_{R}, radial acceleration.

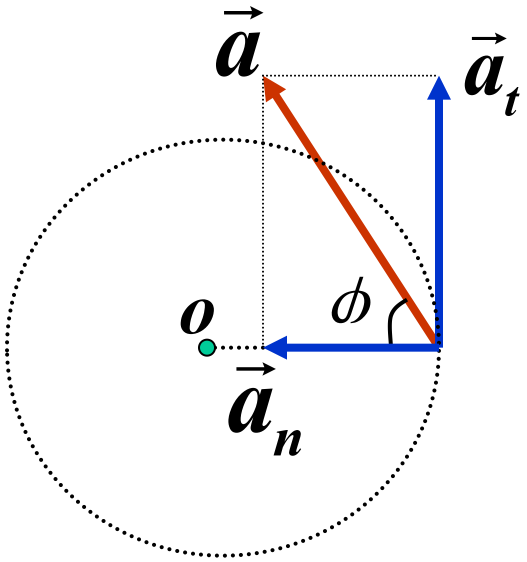
Explanations:

(i) The acceleration in natural coordinate:

$$\boxed{\vec{a} = a_t\vec{\tau} + a_n\vec{n}} \quad \text{where} \quad a_t = \frac{dv}{dt} \qquad a_n = \frac{v^2}{r}$$

or $$a = |\vec{a}| = \sqrt{a_t^2 + a_n^2}$$

$$\varphi = \arctan(a_t / a_n)$$



(ii) For general plane-curve motion:

$$\boxed{\vec{a} = \frac{v^2}{\rho}\vec{n} + \frac{dv}{dt}\vec{\tau}}$$

ρ : radius of curvature of examine position。

Example

Two components of acceleration. A racing car starts from rest in the pit area and accelerates at a uniform rate to a speed of 35 m/s in 11 s, moving on a circular track of radius 500 m. Assuming constant tangential acceleration, find

(a) the tangential acceleration,

(b) the radial acceleration, at the instant when the speed is $v = 15$ m/s, and again when $v = 30$ m/s.

Solution:

(a) the tangential acceleration is constant, of magnitude

$$a_t = \frac{\Delta v}{\Delta t} = \frac{35-0}{11} = 3.2\ \mathrm{m/s^2}$$ and is the same for both instants.

(b) At the earlier instant $a_R = \dfrac{v^2}{r} = \dfrac{15^2}{500} = 0.45\ \mathrm{m/s^2}$

At the second instant $a_R = \dfrac{v^2}{r} = \dfrac{30^2}{500} = 1.8\ \mathrm{m/s^2}$

the radial acceleration continually increases, although the tangential acceleration is constant